

Multiple Choice

1. **Answer:** A

Options: A) 10; B) 100; C) 1,000; D) 10,000

Note: The digit 5 in 24,513 represents 500, while the digit 5 in 357 represents 5, making the value of 5 in 24,513 ten times greater than the value of 5 in 357.

2. **Answer:** B

Options: A) $(5 \times 4) \times (6 \times 5)$; B) $(5 \times 5) + (5 \times 4)$; C) $(5 \times 5) + (5 \times 9)$; D) $(5 \times 9) \times (6 \times 9)$

Note: The expression $(5 \times 5) + (5 \times 4)$ is equivalent to 5×9 because it utilizes the distributive property of multiplication, showing that 9 can be expressed as the sum of 5 and 4.

3. **Answer:** C

Options: A) Small sample size and 95% confidence; B) Small sample size and 99% confidence; C) Large sample size and 95% confidence; D) Large sample size and 99% confidence

Note: A large sample size reduces the standard error, leading to a narrower confidence interval, while a 95% confidence level does not increase the width compared to higher confidence levels, making this combination yield the narrowest interval.

4. **Answer:** C

Options: A) 4; B) 6; C) 10; D) 15

Note: The number must be a multiple of 10 because it is the product of the prime factors 2 and 5, both of which are among the given prime factors.

5. **Answer:** A

Options: A) 6; B) 9; C) 12; D) 14

Note: The correct answer is 6 because the greatest even prime number is 2 and the least odd prime number is 3, and their product (2×3) equals 6.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: The total amount spent on lunch is calculated by adding a 20% tip of 8.80 to the 44 meal, resulting in a total of \$52.80.

7. **Answer:** True

Options: A) True; B) False

Note: The median is the middle value of a data set when arranged in ascending order, and for the set 17.88, 19.88, 21.88, 29.88, 79.88, the middle value is indeed 21.88.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because five strawberries is an odd number, which contradicts the claim of having an even number of strawberries.

9. **Answer:** True

Options: A) True; B) False

Note: There are 31 integers between -200 and 200 that are congruent to 5 modulo 13 because the sequence of integers congruent to 5 modulo 13 can be expressed as $5 + 13k$, where k ranges from -15 to 15, yielding 31 valid integers within the specified range.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the total number of chairs should be calculated by multiplying the number of rows by the number of chairs in each row, not by adding the two numbers together.

Long Form Response

11. **Answer:** We rewrite the given equation as

$$5^{a_{n+1}-a_n} = 1 + \frac{1}{n + \frac{2}{3}} = \frac{3n+5}{3n+2}.$$

Then, we observe a telescoping product:

$$5^{a_n-a_1} = 5^{a_2-a_1} \cdot 5^{a_3-a_2} \cdots 5^{a_n-a_{n-1}} = \frac{8}{5} \cdot \frac{11}{8} \cdots \frac{3n+2}{3n-1} = \frac{3n+2}{5}.$$

Since $a_1 = 1$, we have

$$5^{a_n} = 3n+2$$

for all $n \geq 1$. Thus, a_k is an integer if and only if $3k+2$ is a power of 5. The next power of 5 which is of the form $3k+2$ is $5^3 = 125$, which is $3(41)+2$. Thus $k = 41$.

Note: correct because it transforms the given recurrence relation into a form that allows for a telescoping product, ultimately leading to the conclusion that the sequence defined by the recurrence can be expressed in terms of powers of 5, thereby determining when a_k is an integer based on the condition that $3k+2$ must be a power of 5.

12. **Answer:** To determine the number of basketball teams that can be formed from 30 players, where each team consists of 5 players, we can perform a simple calculation. By dividing the total number of players, 30, by the number of players per team, 5, we find that $30 \div 5 = 6$. This means that we can form 6 teams. Each team will have the same number of players, ensuring that all players are utilized efficiently. Therefore, with 30 players, it is possible to create 6 complete teams.

Note: correct because it accurately divides the total number of players, 30, by the number of players per team, 5, to determine that 6 complete teams can be formed, utilizing all players efficiently.

End of Answer Key